

INTERWAR MILITARY AIRCRAFT

A GENERATION OF MILITARY aircraft developed between the end of WWI and the mid-1930s was obsolete by the time war broke out again in 1939. These were primarily open-cockpit biplanes that achieved better performance than their WWI predecessors – chiefly through improved engines – without any dramatic change in technology. The technical revolution that brought the absolute dominance of streamlined, cantilever-wing monoplanes – often of all-metal construction with retractable undercarriages and closed cockpits – was already apparent in the 1932 Martin B-10B bomber. Some of the biplanes were used in small colonial conflicts, and others saw action in the Spanish Civil War – for example, the Fiat CR.32 and the Nieuport Type 52. A few soldiered on gallantly in the early stages of WWII, including the Gloster Gladiator and Fiat CR.32.

HART FORMATION

Hawker Harts were the RAF's standard light bombers during the 1930s. Their top speed of 184mph (296kph) was considered fast at the time.



Arado Ar 68

The Ar 68 was destined to be the Luftwaffe's last biplane, when it entered service in 1936, replacing the He 51. However, by the time it reached the squadrons, its successor, the Messerschmitt Bf 109 was already being tested.



ENGINE 690hp Junkers Jumo 210 Da

WINGSPAN 36ft 1in (11m) **LENGTH** 31ft 2in (9.5m)

TOP SPEED 208mph (335kph) **CREW** 1

ARMAMENT 2 x 7.92mm MG17s; 132lb (60kg) bombload

Boeing F4B-4



The F4B was Boeing's fourth and most significant member of a family of biplane fighters that had their origins in the 1923 US Army Air Corps' PW-9. The first F4Bs appeared in 1928 and served on the USS *Lexington*. Evolving through a number of variants, the F4B remained in service until the early 1940s.

ENGINE 550hp Pratt & Whitney 9-cylinder radial

WINGSPAN 30ft (9.1m) **LENGTH** 20ft 1in (6.12m)

TOP SPEED 188mph (302kph) **CREW** 1

ARMAMENT 2 x machine guns

Bristol Bulldog IIA

In its Mk.II version, the Bulldog joined the RAF in June 1929. Over 300 were built, equipping ten fighter squadrons and providing 70 percent of Britain's air defenses in the early 1930s. Exports included the air arms of Denmark, Australia, and Finland.



ENGINE 490hp Bristol Jupiter VIIIF 9-cylinder radial

WINGSPAN 33ft 10in (10.3m) **LENGTH** 25ft 2in (7.7m)

TOP SPEED 174mph (280kph) **CREW** 1

ARMAMENT 2 x machine guns; 80lb (35kg) bombload

Dewoitine D.500

All-metal construction

Fixed undercarriage

The D.500 joined the French air force in 1935. Despite its open cockpit and fixed undercarriage, the D.500 provided the Armée de L'Air with its first modern all-metal monoplane fighter. The fighter was so important that production was shared between three manufacturers, including Dewoitine. The D.500, or its cannon-armed variant the D.501, were also exported to China, Lithuania, and Venezuela. A total of 308 D.500/501s were constructed, and the last 501s were withdrawn from service in 1941.



ENGINE 600hp Hispano-Suiza 12Xbrs V-12

WINGSPAN 39ft 8in (12m) **LENGTH** 25ft 5in (7.7m)

TOP SPEED 223mph (359kph) **CREW** 1

ARMAMENT 4 x machine guns

Farman F.222

A four-engined bomber was a rare sight during the 1930s, the Farman F.220 series and Tupolev TB-3 among the few. Farman's were initially used in night bombing raids over Germany, before being converted to military transports (civilian version shown here).



ENGINE 4 x 950hp Gnome-Rhone 14N 14-cylinder radials

WINGSPAN 118ft 1in (36m) **LENGTH** 70ft 5in (21.4m)

TOP SPEED 199mph (320kph) **CREW** 5

ARMAMENT 3 x machine guns; 9,260lb (4,200kg) bombload